

Healthy Bite

Complete Relational Database Data Dictionary (16 Tables - 3NF)

Database Name: healthy_bite | **Engine:** InnoDB | **Charset:** utf8mb4_unicode_ci
Architecture: Multi-Tenant QR Ordering (3NF) | **Total Tables:** 16 | **Primary Key Standard:** BIGINT UNSIGNED AUTO_INCREMENT

System Database Overview

The **Healthy Bite** relational database architecture comprises 16 normalized tables designed for real-time QR code table ordering, multi-branch tenant management, dynamic digital menus, nutritional/allergen logging, order processing, and payment settlement. Strict foreign key constraints ensure referential integrity across all transactional entities with cascading updates and deletes.

1. admin

System administrator entities with platform-wide management permissions.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique identifier for platform super admin.
name	VARCHAR(120)	No	None	NULL	Full display name of administrator.
created_at	TIMESTAMP	No	None	CURRENT_TIMES TAMP	Record creation timestamp.
updated_at	TIMESTAMP	No	None	CURRENT_TIMES TAMP	Timestamp of last administrative update.

2. users

User authentication accounts linked to managing administrators and restaurant tenants.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique user account ID.
admin_id	BIGINT UNSIGNED	No	FK (admin.id)	NULL	Foreign key referencing assigned managing admin.
restaurant_id	BIGINT UNSIGNED	Yes	FK (restaurants.id)	NULL	Foreign key linking user to restaurant tenant.
name	VARCHAR(120)	No	None	NULL	Full user display name.
email	VARCHAR(190)	No	UNIQUE INDEX	NULL	Unique user email address for login.
password_hash	VARCHAR(255)	No	None	NULL	Bcrypt hashed password credential.
status	ENUM('active', 'inactive')	No	None	'active'	Account activity state.
created_at	TIMESTAMP	No	None	CURRENT_TIMES TAMP	Account creation timestamp.
updated_at	TIMESTAMP	No	None	CURRENT_TIMES TAMP	Profile update timestamp.

3. restaurants

Restaurant tenant establishments, business profiles, contact info, and approval statuses.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique restaurant organization ID.
owner_user_id	BIGINT UNSIGNED	Yes	FK (users.id)	NULL	Foreign key linking restaurant to owner user account.
name	VARCHAR(160)	No	None	NULL	Brand or establishment name.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
email	<code>VARCHAR(190)</code>	Yes	None	NULL	Business contact email address.
phone	<code>VARCHAR(30)</code>	Yes	None	NULL	Business telephone number.
address	<code>VARCHAR(500)</code>	Yes	None	NULL	Street address of primary establishment.
city	<code>VARCHAR(120)</code>	Yes	None	NULL	City location.
state	<code>VARCHAR(120)</code>	Yes	None	NULL	State or province location.
cuisine_type	<code>VARCHAR(120)</code>	Yes	None	NULL	Cuisine classification (e.g. Italian, Multi-Cuisine).
description	<code>VARCHAR(1000)</code>	Yes	None	NULL	Detailed promotional overview text.
approval_status	<code>ENUM('pending', 'approved', 'suspended')</code>	No	None	'approved'	Platform verification state.
created_at	<code>TIMESTAMP</code>	No	None	<code>CURRENT_TIMESTAMP</code>	Organization creation timestamp.
updated_at	<code>TIMESTAMP</code>	No	None	<code>CURRENT_TIMESTAMP</code>	Modification timestamp.

4. branches

Specific branch outlets operated under a parent restaurant organization.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	<code>BIGINT UNSIGNED</code>	No	PRIMARY (AI)	NULL	Unique branch outlet ID.
restaurant_id	<code>BIGINT UNSIGNED</code>	No	FK (restaurants.id)	NULL	Foreign key linking branch to parent restaurant.
name	<code>VARCHAR(160)</code>	No	None	NULL	Branch outlet location name (e.g. Downtown Branch).
phone	<code>VARCHAR(30)</code>	Yes	None	NULL	Branch direct phone number.
address	<code>VARCHAR(500)</code>	Yes	None	NULL	Physical address of outlet.
created_at	<code>TIMESTAMP</code>	No	None	<code>CURRENT_TIMESTAMP</code>	Branch creation timestamp.
updated_at	<code>TIMESTAMP</code>	No	None	<code>CURRENT_TIMESTAMP</code>	Branch update timestamp.

5. categories

Digital menu category groupings (e.g., Starters, Salads, Beverages) per restaurant.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	<code>BIGINT UNSIGNED</code>	No	PRIMARY (AI)	NULL	Unique category ID.
restaurant_id	<code>BIGINT UNSIGNED</code>	No	FK (restaurants.id)	NULL	Foreign key linking category to restaurant.
name	<code>VARCHAR(120)</code>	No	UNIQUE (rest, name)	NULL	Category display name.
sort_order	<code>INT UNSIGNED</code>	No	None	0	Display order sequence index on menu UI.
is_active	<code>TINYINT(1)</code>	No	None	1	Active menu visibility flag (1 = Active, 0 = Hidden).
created_at	<code>TIMESTAMP</code>	No	None	<code>CURRENT_TIMESTAMP</code>	Category creation timestamp.
updated_at	<code>TIMESTAMP</code>	No	None	<code>CURRENT_TIMESTAMP</code>	Category update timestamp.

6. food_items

Dishes and food items, nutritional metrics, prices, allergen tags, and stock availability.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique food item ID.
category_id	BIGINT UNSIGNED	No	FK (categories.id)	NULL	Foreign key linking dish to menu category.
name	VARCHAR(160)	No	None	NULL	Dish name.
ingredients	TEXT	Yes	None	NULL	Ingredients list.
base_price	DECIMAL(10, 2)	No	None	NULL	Base item price.
image	VARCHAR(255)	Yes	None	NULL	Image relative path or URL.
calories	INT	Yes	None	NULL	Calorie count (kcal).
protein	INT	Yes	None	NULL	Protein content in grams.
carbs	INT	Yes	None	NULL	Carbohydrates in grams.
fat	INT	Yes	None	NULL	Fat content in grams.
allergens	VARCHAR(255)	Yes	None	NULL	Allergen warnings (e.g. Nuts, Gluten).
preparation_time	INT	Yes	None	NULL	Estimated cooking time in minutes.
spice_level	VARCHAR(50)	Yes	None	NULL	Spice rating (e.g., Mild, Medium, Hot).
food_type	VARCHAR(50)	No	None	'veg'	Classification (veg, non-veg, vegan, egg).
is_available	BOOLEAN	No	None	1	In-stock availability toggle.
is_featured	BOOLEAN	No	None	0	Featured special item flag.
created_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Item creation timestamp.
updated_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Item update timestamp.

7. restaurant_tables

Physical dining tables situated at branch locations with occupancy statuses.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique dining table ID.
branch_id	BIGINT UNSIGNED	No	FK (branches.id)	NULL	Foreign key linking table to branch location.
table_number	VARCHAR(80)	No	UNIQUE (branch, num)	NULL	Table designation code/number (e.g. T-01).
status	ENUM('available', 'occupied', 'cleaning', 'out_of_service')	No	None	'available'	Real-time table state.
created_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Table creation timestamp.
updated_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Table update timestamp.

8. qr_tokens

Cryptographic QR session tokens assigned to physical dining tables for instant mobile ordering.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique QR token session ID.
restaurant_table_id	BIGINT UNSIGNED	No	FK (restaurant_tables.id)	NULL	Foreign key linking token to physical table.
token	VARCHAR(255)	No	UNIQUE INDEX	NULL	Cryptographic token string.
expires_at	DATETIME	Yes	None	NULL	Session expiration timestamp.
is_active	TINYINT(1)	No	None	1	Active session validity flag.
created_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Token creation timestamp.
updated_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Token update timestamp.

9. food_variants

Portion size variants for food items (e.g. Small, Medium, Large) with price adjustments.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique variant ID.
food_item_id	BIGINT UNSIGNED	No	FK (food_items.id)	NULL	Foreign key linking variant to food item.
name	VARCHAR(80)	No	None	NULL	Variant name (e.g. Half, Full, 500ml).
price_adjustment	DECIMAL(10,2)	No	None	0.00	Price difference relative to base price.
created_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Variant creation timestamp.
updated_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Variant update timestamp.

10. food_customizations

Custom add-on options and toppings (e.g., Extra Cheese, Low Salt) with pricing.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique customization option ID.
food_item_id	BIGINT UNSIGNED	No	FK (food_items.id)	NULL	Foreign key linking add-on to food item.
name	VARCHAR(120)	No	None	NULL	Add-on name (e.g., Extra Dip, Extra Cheese).
price_adjustment	DECIMAL(10,2)	No	None	0.00	Additional price for customization.
created_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Add-on creation timestamp.
updated_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Add-on update timestamp.

11. customers

Customer profiles, guest names, and mobile numbers for digital receipts and order tracking.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique customer profile ID.
name	VARCHAR(120)	No	None	NULL	Guest customer name.
phone	VARCHAR(30)	Yes	None	NULL	Contact telephone number.
created_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Profile creation timestamp.
updated_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Profile update timestamp.

12. reviews

Ratings and text comments submitted by customers for dishes and dining experiences.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique review record ID.
customer_id	BIGINT UNSIGNED	No	FK (customers.id)	NULL	Foreign key referencing reviewing customer.
restaurant_id	BIGINT UNSIGNED	No	FK (restaurants.id)	NULL	Foreign key referencing reviewed restaurant.
food_item_id	BIGINT UNSIGNED	Yes	FK (food_items.id)	NULL	Optional dish reference for item reviews.
restaurant_table_id	BIGINT UNSIGNED	Yes	FK (restaurant_tables.id)	NULL	Optional table location reference.
rating	INT	No	None	NULL	Star rating score (1 to 5).
comment	TEXT	Yes	None	NULL	Customer feedback text.
created_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Review submission timestamp.
updated_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Review update timestamp.

13. orders

Customer table orders, unique tracking order numbers, status workflow, and bill subtotal/total.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique order transaction ID.
branch_id	BIGINT UNSIGNED	No	FK (branches.id)	NULL	Foreign key referencing branch outlet.
customer_id	BIGINT UNSIGNED	No	FK (customers.id)	NULL	Foreign key referencing customer.
restaurant_table_id	BIGINT UNSIGNED	No	FK (restaurant_tables.id)	NULL	Foreign key referencing dining table.
order_number	VARCHAR(32)	No	UNIQUE INDEX	NULL	Unique tracking order number string.
status	ENUM('pending', 'accepted', 'preparing', 'ready', 'served', 'completed', 'cancelled')	No	INDEX	'pending'	Kitchen order status.
customer_note	VARCHAR(500)	Yes	None	NULL	Special instructions from guest.
subtotal	DECIMAL(10,2)	No	None	NULL	Order subtotal before tax/discounts.
total_amount	DECIMAL(10,2)	No	None	NULL	Final total bill payable.
created_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Order placement timestamp.
updated_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Order status update timestamp.

14. order_items

Individual line items associated with an order, item snapshots, quantities, and line totals.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique order line item ID.
order_id	BIGINT UNSIGNED	No	FK (orders.id)	NULL	Foreign key referencing parent order.
food_item_id	BIGINT UNSIGNED	No	FK (food_items.id)	NULL	Foreign key linking item to menu.
food_variant_id	BIGINT UNSIGNED	Yes	FK (food_variants.id)	NULL	Selected variant portion size reference.
item_name	VARCHAR(160)	No	None	NULL	Historical snapshot of item name.
unit_price	DECIMAL(10,2)	No	None	NULL	Unit price charged.
quantity	INT	No	None	NULL	Item quantity ordered.
line_total	DECIMAL(10,2)	No	None	NULL	Calculated line total (unit_price * quantity).
created_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Line item creation timestamp.

15. order_item_customizations

Pivot mapping tracking add-ons and customizations chosen for each ordered line item.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique line item customization ID.
order_item_id	BIGINT UNSIGNED	No	FK (order_items.id)	NULL	Foreign key linking customization to order line item.
food_customization_id	BIGINT UNSIGNED	No	FK (food_customizations.id)	NULL	Foreign key referencing selected add-on.
created_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Selection timestamp.

16. payments

Payment transaction records, settlement channels (cash, UPI, card), and payment verification state.

Attribute Name	Data Type	Null	Key / Constraint	Default Value	Description
id	BIGINT UNSIGNED	No	PRIMARY (AI)	NULL	Unique payment transaction ID.
order_id	BIGINT UNSIGNED	No	FK (orders.id)	NULL	Foreign key linking payment to settled order.
amount	DECIMAL(10,2)	No	None	NULL	Paid monetary amount.
method	ENUM('cash', 'upi', 'card')	No	None	NULL	Payment channel method.
status	ENUM('pending', 'completed', 'failed')	No	None	'pending'	Payment status.
created_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Payment timestamp.
updated_at	TIMESTAMP	No	None	CURRENT_TIMESTAMP	Payment update timestamp.